CLACELL: Robust cell type classifier for immune cells

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Motivation

- Cell type annotation is a critical prerequesite for downstream biological insights (e.g. gene regulatory network modeling)
- Existing cell type classifiers suffer from a lack of robustness, low F1-Scores, high variance and sensitivity to gene dropout

=> CLACELL: a Python package designed for the robust and deterministic classification of immune cells

Training

Annotate Datasets for training
using MarkerAnnotator

Preprocess datasets
using preprocess_data()

Select Classifier C
CellClassifier or
ConditionalCellClassifier

Train Model
using C.random_search(),
C.bayes_search() or C.train()

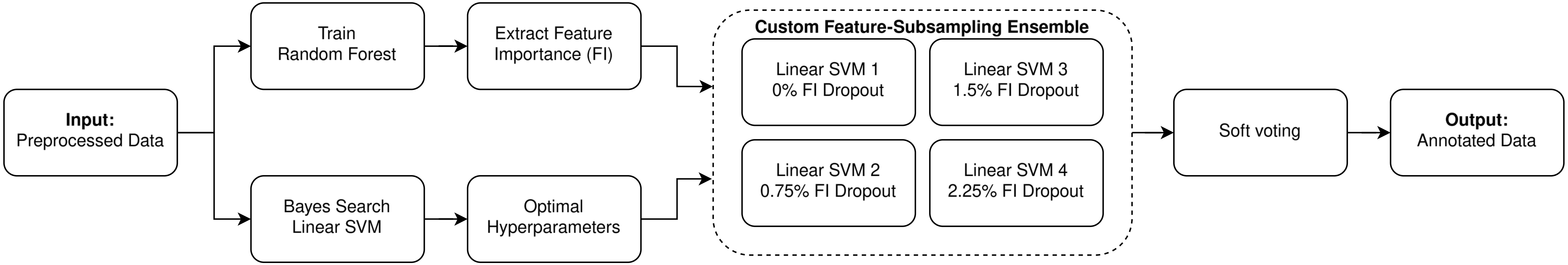
Evaluate Model
using C.evaluate()

Inference

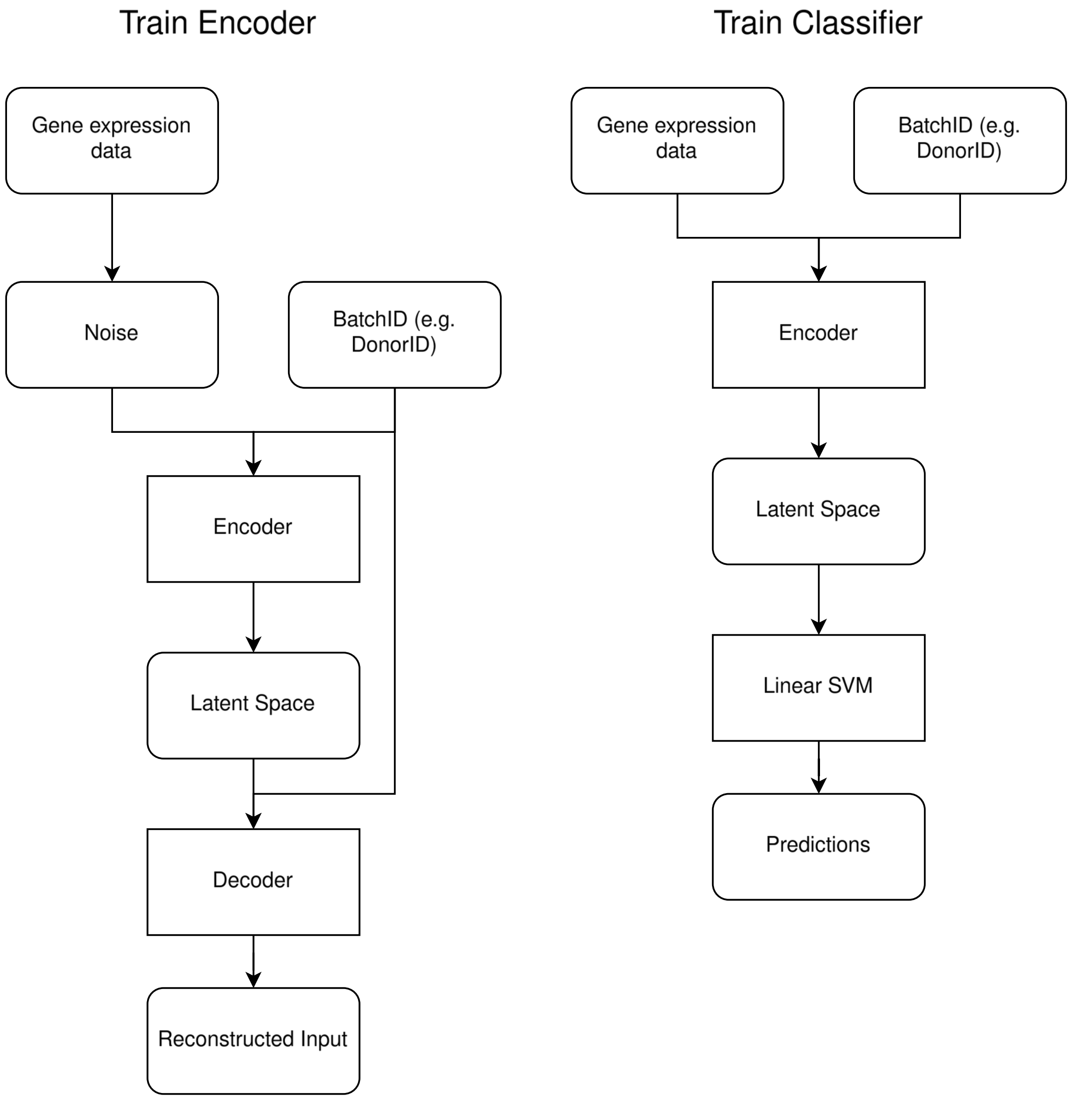
Preprocess datasets
using preprocess_data()

Annotate Datasets
using C.predict() on a
trained Model C

CellClassifier: Custom Ensemble

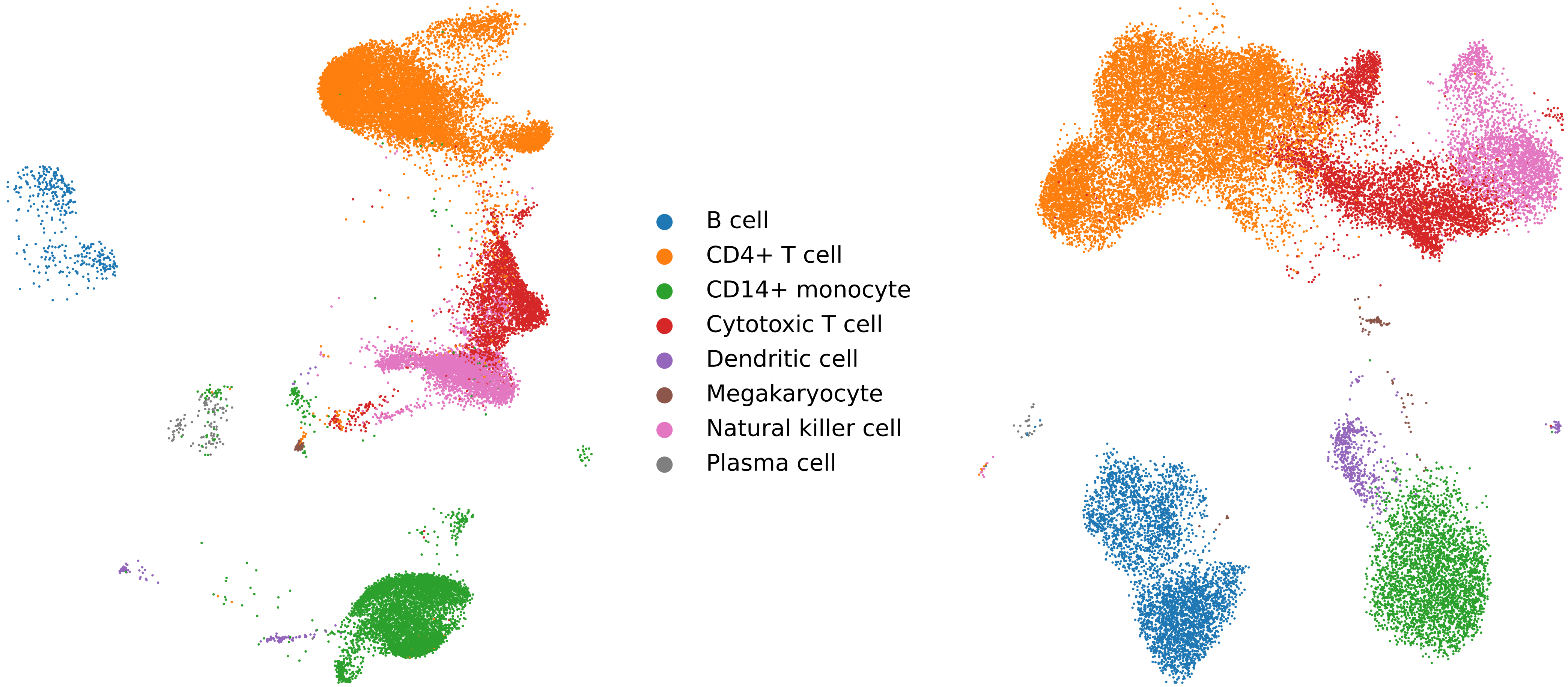


ConditionalCellClassifier: Conditional Autoencoder



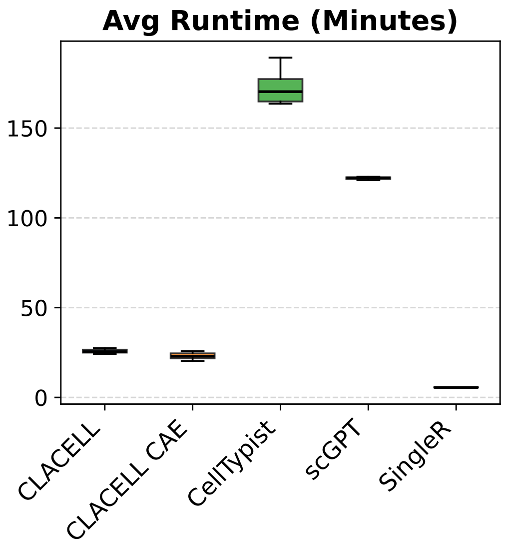
In-Distribution and Out-of-Distribution Datasets

In-Distribution (ID) | Out-of-Distribution (OOD)



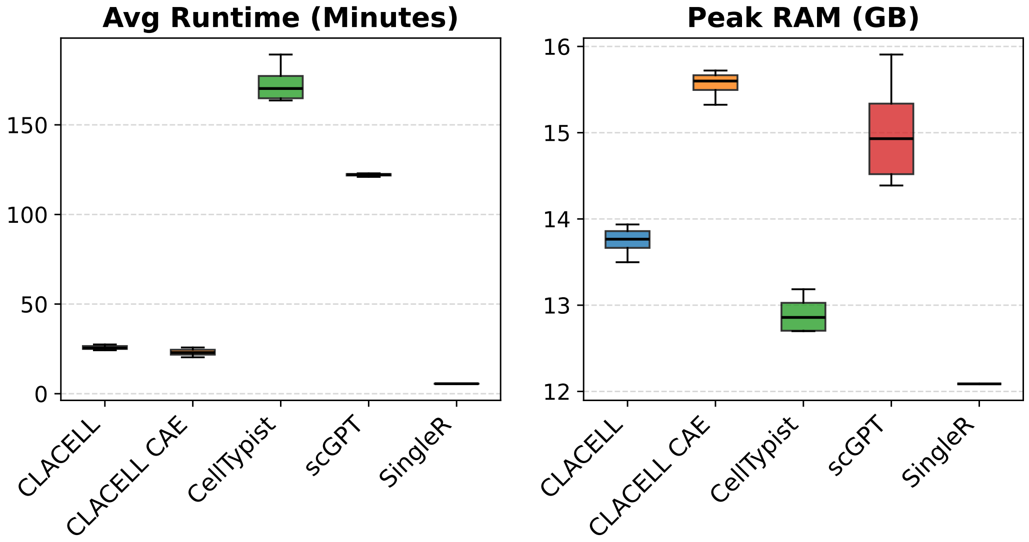
Results

Overall Comparison



Results

Overall Comparison



Results

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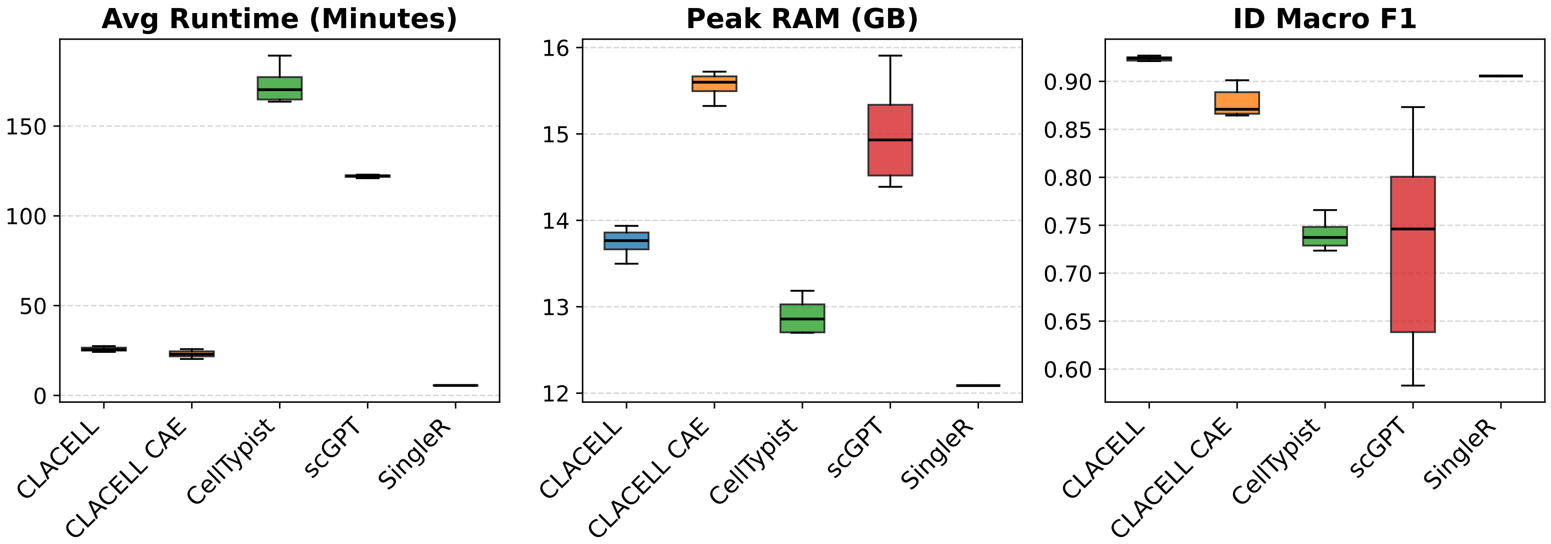

Results

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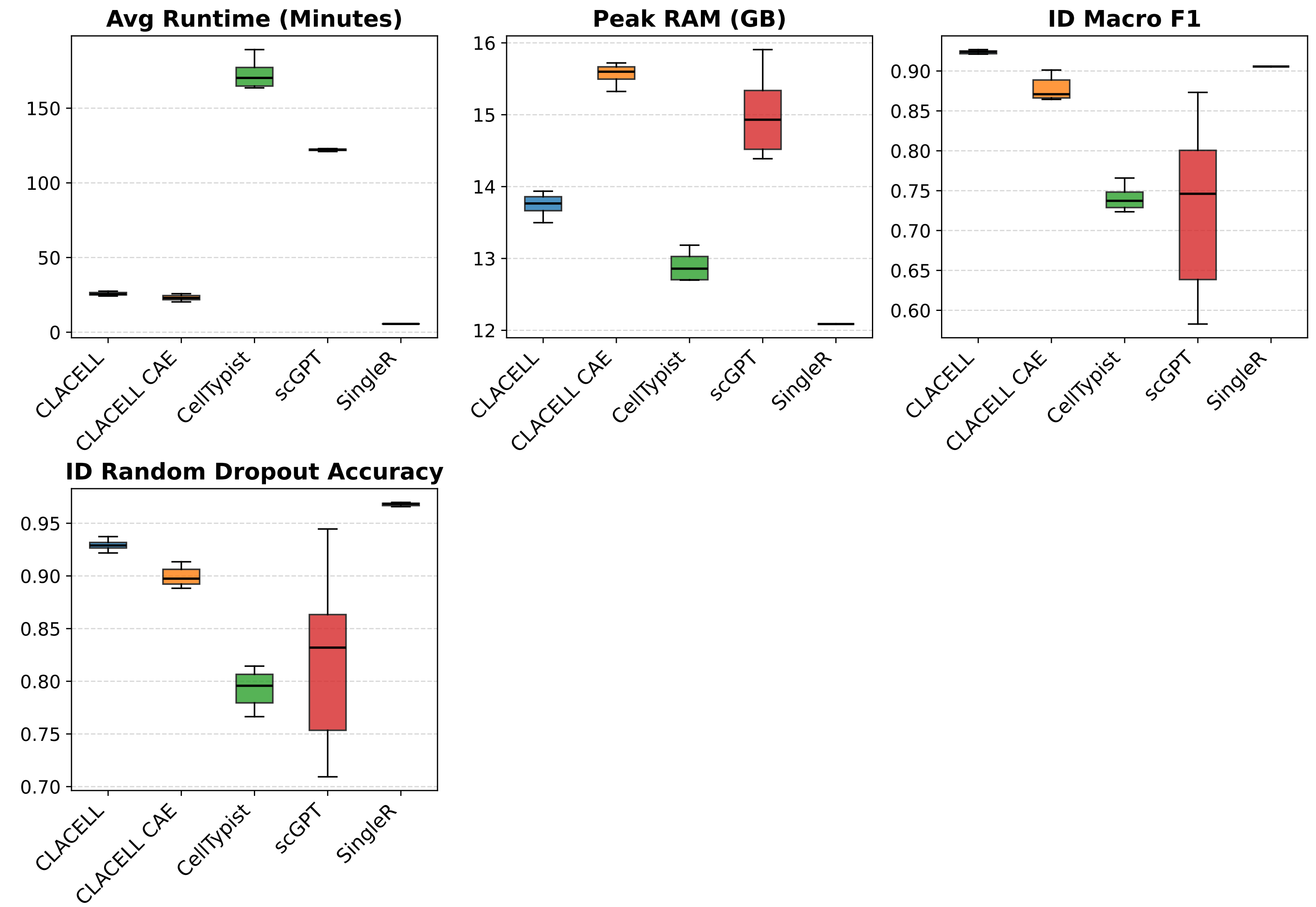
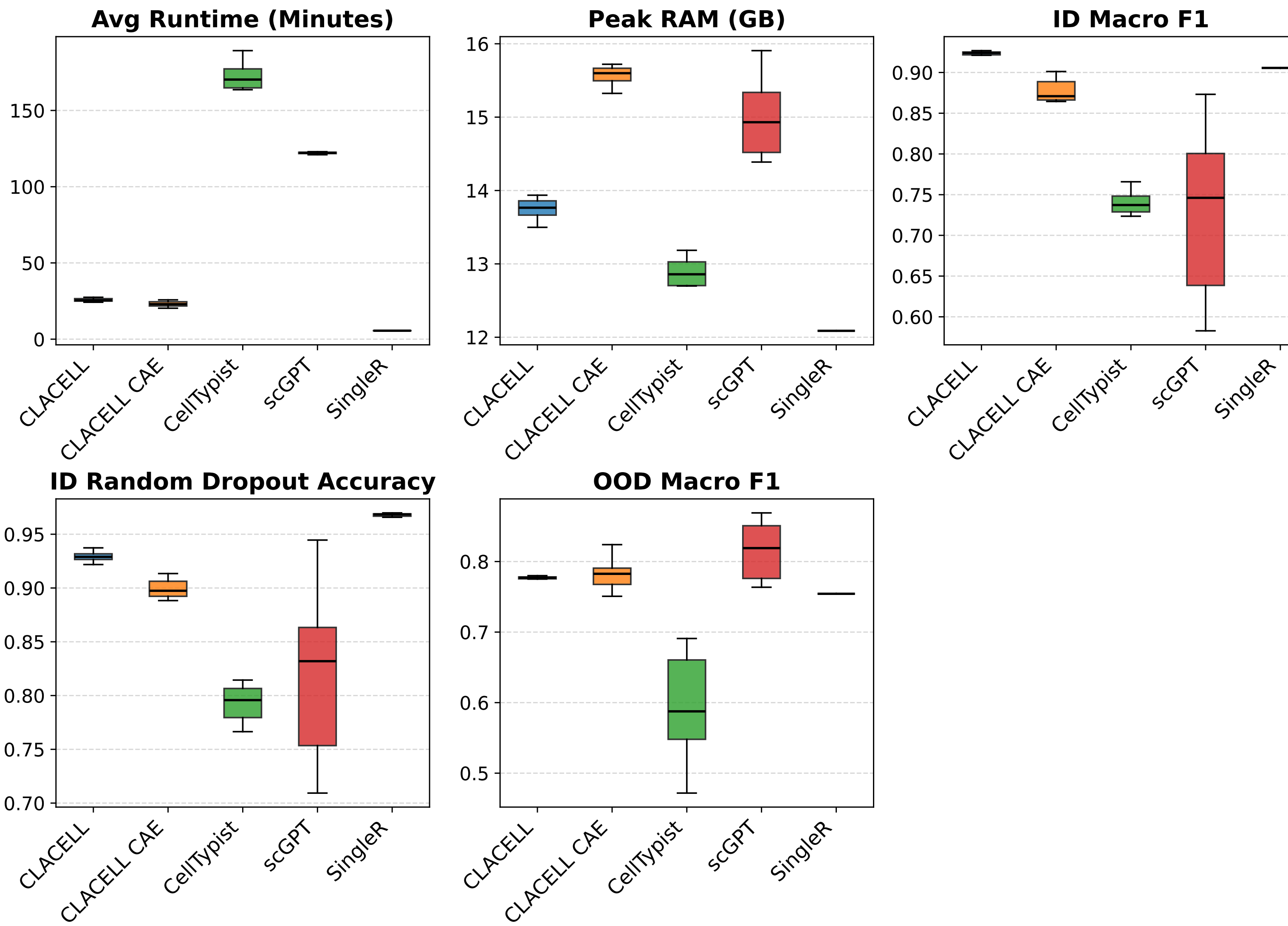


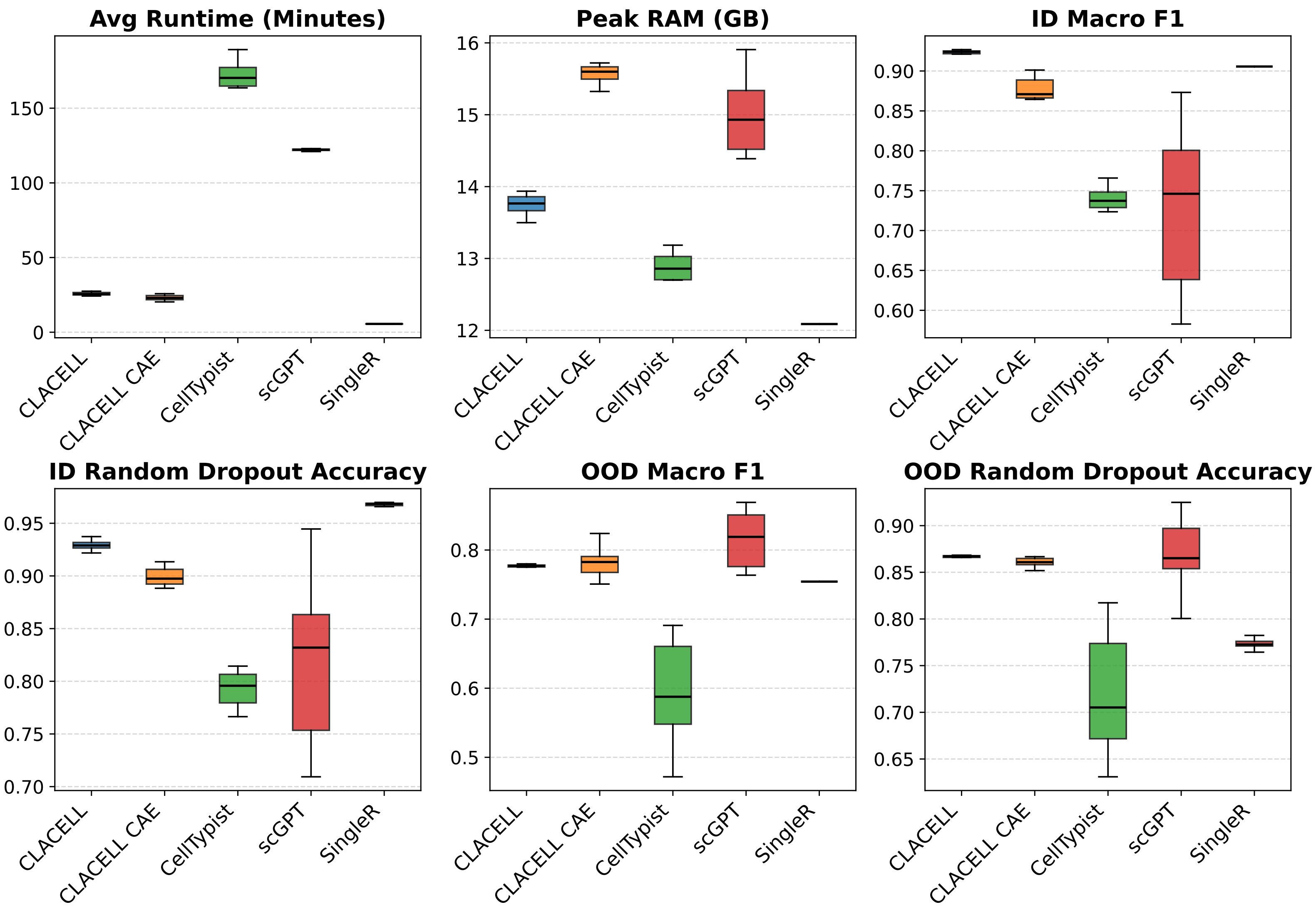
Results

Overall Comparison



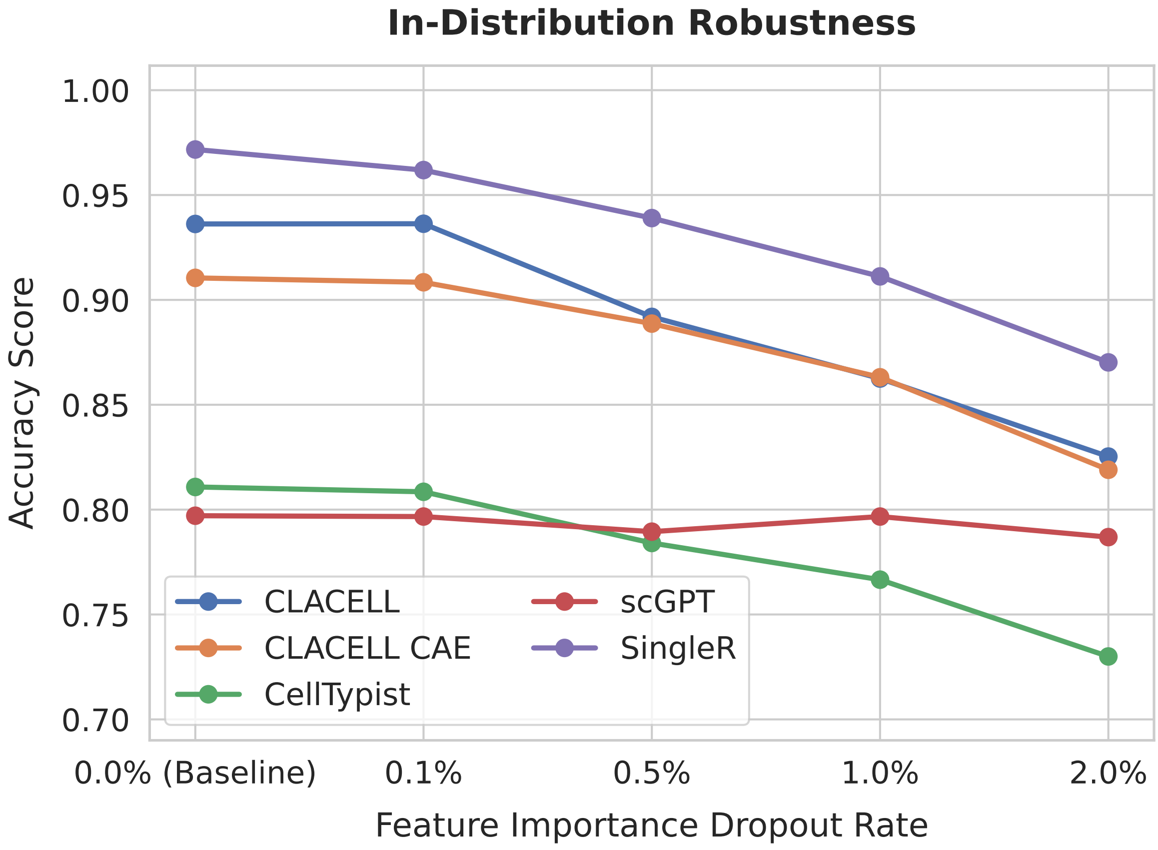
Results

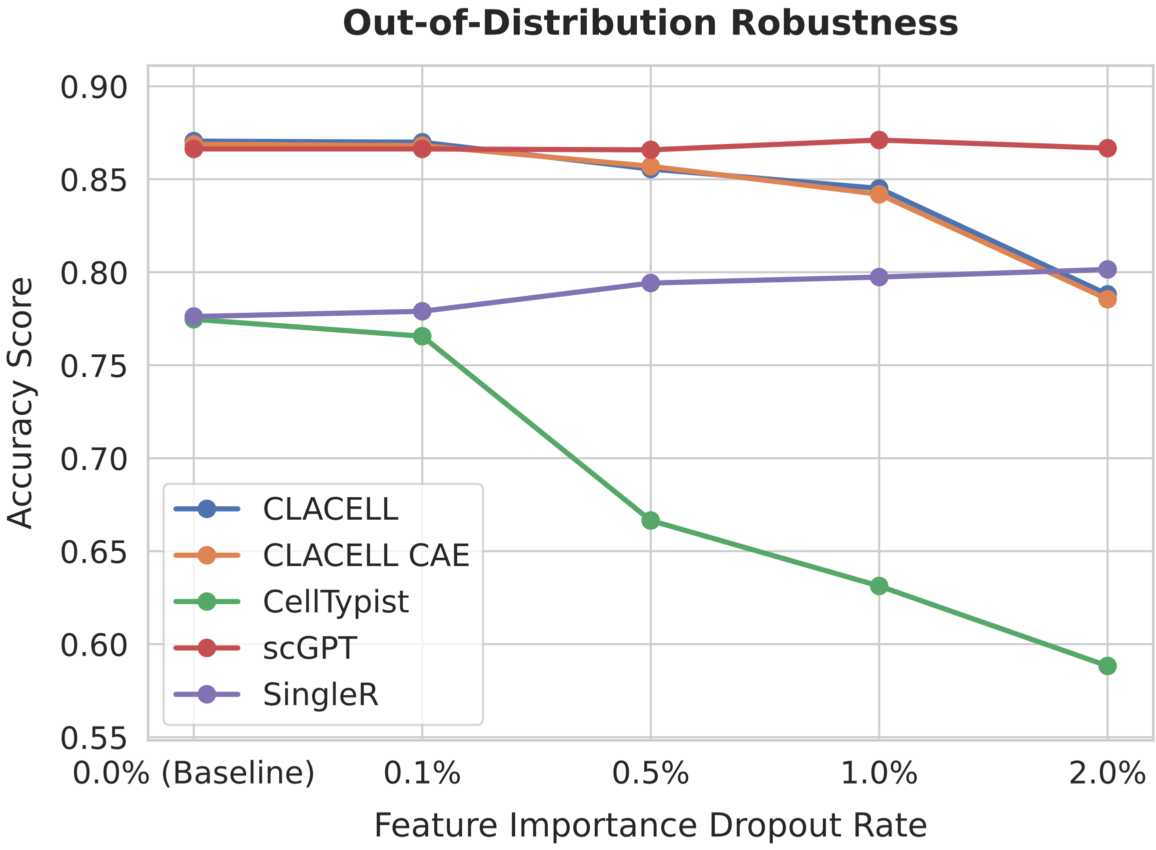
Overall Comparison



Results

Feature Importance Dropout





Conclusion

- Python Package for the robust and deterministic annotation of immune cells
- CellClassifier offers a lower standard deviation, ConditionalCellClassifier generalizes slightly better on out-of-distribution data
- Future Work:
  - classify more fine-grained cell types
  - evaluate Package on spatial data

Code: https://github.com/Boolean-true/CLACELL